

► **Example 5.7.34** Solve by the method of variation of parameters:

$$\frac{d^2 y}{dx^2} + a^2 y = \sec ax.$$

[C.H. 1993, '95, '99, 2001]

Solution: To find C.F.

Auxiliary equation is $m^2 + a^2 = 0$ which gives $m = \pm ia$.

The C.F. is $c_1 \cos ax + c_2 \sin ax$.

To find P.I.

First show that $\cos ax$ and $\sin ax$ are independent solutions.

$$\text{Let } y_1 = \cos ax, \quad y_2 = \sin ax.$$

The Wronskian of y_1 and y_2 is

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \cos ax & \sin ax \\ -a \sin ax & a \cos ax \end{vmatrix} = a \neq 0.$$

Therefore, y_1 and y_2 are independent solutions.

Let the particular solution be

$$y_p = v_1(x) \cos ax + v_2(x) \sin ax \quad (5.7.33)$$

$$\begin{aligned} \therefore Dy_p &= \{-av_1(x) \sin ax + a v_2(x) \cos ax\} \\ &\quad + \{v_1'(x) \cos ax + v_2'(x) \sin ax\}. \end{aligned}$$

Choose $v_1(x)$ and $v_2(x)$ such that

$$v_1'(x) \cos ax + v_2'(x) \sin ax = 0 \quad (5.7.34)$$

so that

$$\begin{aligned} Dy_p &= -av_1(x) \sin ax + av_2(x) \cos ax \\ \text{and } D^2 y_p &= \{-a^2 v_1(x) \cos ax - a^2 v_2(x) \sin ax\} \\ &\quad - av_1'(x) \sin ax + av_2'(x) \cos ax. \end{aligned}$$

Now the given equation is

$$\begin{aligned} D^2 y_p + a^2 y_p &= \sec ax \\ \text{or, } -a^2 v_1(x) \cos ax - a^2 v_2(x) \sin ax - av_1'(x) \sin ax + av_2'(x) \cos ax \\ &\quad + a^2 v_1(x) \cos ax + a^2 v_2(x) \sin ax = \sec ax \\ \text{or, } -av_1'(x) \sin ax + av_2'(x) \cos ax &= \sec ax. \end{aligned} \quad (5.7.35)$$

From (5.7.33) and (5.7.35)

$$v_1'(x) \cos ax + v_2'(x) \sin ax = 0$$

$$-av_1'(x) \sin ax + av_2'(x) \cos ax = \sec ax.$$

Solving

$$v_1'(x) = -\frac{1}{a} \tan ax,$$

$$v_2'(x) = \frac{1}{a}.$$

$$\therefore v_1(x) = -\frac{1}{a} \int \tan ax \, dx = -\frac{1}{a^2} \log(\sec ax)$$

$$\text{and } v_2(x) = \frac{1}{a} \int dx = \frac{x}{a}.$$

Accordingly,

$$y_p = -\frac{1}{a^2} \cos ax \log(\sec ax) + \frac{1}{a} x \sin ax.$$

The complete solution is therefore

$$y = c_1 \cos ax + c_2 \sin ax - \frac{1}{a^2} \cos ax \log(\sec ax) + \frac{1}{a} x \sin ax.$$

► **Example 5.7.35** Solve $(D^2 - 2D)y = e^x \cos x$ by the method of variation of parameters.

Solution: C.F. is $c_1 + c_2 e^{2x}$ (c_1, c_2 arbitrary constants).

We take $y_1(x) = 1$ and $y_2(x) = e^{2x}$ and take $y_p = v_1(x) + v_2(x)e^{2x}$ (Replacing c_1 and c_2 by $v_1(x)$ and $v_2(x)$). $y_1(x)$ and $y_2(x)$ are linearly independent, since

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} 1 & e^{2x} \\ 0 & 2e^{2x} \end{vmatrix} = 2e^{2x} \neq 0.$$

Now

$$Dy_p = v_1'(x) + v_2'(x)e^{2x} + 2v_2(x)e^{2x}.$$

Choose $v_1(x)$ and $v_2(x)$ such that

$$v_1'(x) + v_2'(x)e^{2x} = 0.$$

Then

$$\begin{aligned}
 Dy_p &= 2v_2(x)e^{2x} \\
 \text{and } D^2y_p &= 2v_2'(x)e^{2x} + 4v_2(x)e^{2x} \\
 (D^2 - 2D)y_p &= 2v_2'(x)e^{2x} + 4v_2(x)e^{2x} - 4v_2(x)e^{2x} \\
 &= 2v_2'(x)e^{2x} = e^x \cos x \\
 \therefore v_2'(x) &= \frac{1}{2}e^{-x} \cos x. \\
 \therefore v_2(x) &= \frac{1}{2} \int e^{-x} \cos x \, dx = \frac{1}{4}e^{-x}(-\cos x + \sin x).
 \end{aligned}$$

Now

$$\begin{aligned}
 v_1'(x) &= -v_2'(x)e^{2x} = -\frac{1}{2}e^x \cos x \\
 \therefore v_1(x) &= -\frac{1}{4}e^x(\cos x + \sin x)
 \end{aligned}$$

$\therefore$ a particular solution of the given equation is

$$\begin{aligned}
 y_p &= v_1(x) + v_2(x)e^{2x} \\
 &= -\frac{1}{4}e^x(\cos x + \sin x) + \frac{1}{4}(\sin x - \cos x)e^x \\
 &= -\frac{1}{2}e^x \cos x
 \end{aligned}$$

and the primitive is

$$y = c_1 + c_2e^{2x} - \frac{1}{2}e^x \cos x.$$

► **Example 5.7.36** Solve by the method of variation of parameters:

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + y = \frac{e^{-x}}{x^2}.$$

[C.H. 1996, 2002]

Solution: To find C.F.

Auxiliary equation is

$$m^2 + 2m + 1 = 0 = (m + 1)^2$$

whereby $m = -1, -1$.

$$\text{C.F. is } e^{-x}(c_1 + c_2x) = c_1e^{-x} + c_2xe^{-x}.$$

To find P.I.

First to show that e^{-x} and xe^{-x} are independent solutions.

$$\text{Let } y_1 = e^{-x}, \quad y_2 = xe^{-x}.$$

The Wronskian of y_1 and y_2 is

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^{-x} & xe^{-x} \\ -e^{-x} & e^{-x}(1-x) \end{vmatrix} = e^{-2x} \neq 0.$$

Therefore, y_1 and y_2 are independent solutions.

Let the particular solution be

$$\begin{aligned} y_p &= v_1(x)e^{-x} + v_2(x)xe^{-x}. \\ \therefore Dy_p &= \{-v_1(x)e^{-x} + (1-x)v_2(x)e^{-x}\} \\ &\quad + \{v_1'(x)e^{-x} + v_2'(x)xe^{-x}\}. \end{aligned} \quad (5.7.36)$$

Choose $v_1(x)$ and $v_2(x)$ such that

$$v_1'(x)e^{-x} + xv_2'(x)e^{-x} = 0 \quad (5.7.37)$$

so that

$$\begin{aligned} Dy_p &= -v_1(x)e^{-x} + (1-x)v_2(x)e^{-x} \\ \text{and } D^2y_p &= \{v_1(x)e^{-x} + (x-2)v_2(x)e^{-x}\} \\ &\quad -v_1'(x)e^{-x} + (1-x)v_2'(x)e^{-x}. \end{aligned}$$

Now the given equation is

$$\begin{aligned} D^2y_p + 2Dy_p + y_p &= \frac{e^{-x}}{x^2} \\ \text{or, } v_1(x)e^{-x} + (x-2)v_2(x)e^{-x} - v_1'(x)e^{-x} + (1-x)v_2'(x)e^{-x} \\ &\quad - 2v_1(x)e^{-x} + 2(1-x)v_2(x)e^{-x} + v_1(x)e^{-x} \\ &\quad + v_2(x)xe^{-x} = \frac{e^{-x}}{x^2} \\ \text{or, } -v_1'(x)e^{-x} + (1-x)v_2'(x)e^{-x} &= \frac{e^{-x}}{x^2}. \end{aligned} \quad (5.7.38)$$

From (5.7.37) and (5.7.38)

$$\begin{aligned} v_1'(x)e^{-x} + xv_2'(x)e^{-x} &= 0 \\ -v_1'(x)e^{-x} + (1-x)v_2'(x)e^{-x} &= \frac{e^{-x}}{x^2}. \end{aligned}$$

Solving

$$v_1'(x) = -\frac{1}{x}, \quad v_2'(x) = \frac{1}{x^2}$$

whereby $v_1(x) = -\log x$, $v_2(x) = -\frac{1}{x}$.

Accordingly

$$y_p = -e^{-x} \log x - e^{-x} = -e^{-x}(1 + \log x)$$

Complete solution is $y = e^{-x}(c_1 + c_2 x) - e^{-x}(1 + \log x)$.

► **Example 5.7.37** Solve by the method of variation of parameters:

$$\frac{d^2 y}{dx^2} + a^2 y = \tan ax.$$

[C.H. 1998]

Solution: Proceed exactly as in Ex. 5.7.34.

To find C.F.

C.F. is $c_1 \cos ax + c_2 \sin ax$.

To find P.I.

Call $y_1 = \cos ax$, $y_2 = \sin ax$ and it has been observed that y_1, y_2 are independent solutions. Also the particular solution will be of the form

$$y_p = v_1(x) \cos ax + v_2(x) \sin ax.$$

Next proceed as before where $v_1'(x), v_2'(x)$ can be found from

$$v_1' \cos ax + v_2' \sin ax = 0$$

$$\text{and } -av_1' \sin ax + av_2' \cos ax = \tan ax.$$

Solving

$$v_1'(x) = -\frac{1}{a} \frac{\sin^2 ax}{\cos ax}$$

$$v_2'(x) = \frac{1}{a} \sin ax.$$

$$v_1(x) = -\frac{1}{a} \int (\sec ax - \cos ax) dx$$

$$= -\frac{1}{a^2} \log |\sec ax + \tan ax| + \frac{1}{a^2} \sin ax$$

$$\text{and } v_2(x) = \frac{1}{a} \int \sin ax dx = -\frac{1}{a^2} \cos ax.$$

$$\therefore y_p = -\frac{1}{a^2} \cos ax \log |\sec ax + \tan ax|.$$

Complete solution is $y = c_1 \cos ax + c_2 \sin ax + y_p$.

► **Example 5.7.38** Solve by the method of variation of parameters:

$$\frac{d^3 y}{dx^3} + \frac{dy}{dx} = \operatorname{cosec} x.$$

Solution: To find C.F.

Auxiliary equation $m^3 + m = 0 = m(m+i)(m-i)$, then $m = 0, \pm i$.

C.F. is $c_1 + c_2 \cos x + c_3 \sin x$.

To find P.I.

Let the particular solution y_p be

$$y_p = v_1(x) + v_2(x) \cos x + v_3(x) \sin x$$

[replace c_1, c_2, c_3 by v_1, v_2, v_3 respectively] where $v_1(x), v_2(x), v_3(x)$ are unknown functions.

Now v'_1, v'_2, v'_3 are found by solving

$$\begin{aligned} v'_1(1) + v'_2(\cos x) + v'_3(\sin x) &= 0 \\ v'_1(0) + v'_2(-\sin x) + v'_3(\cos x) &= 0 \\ v'_1(0) + v'_2(-\cos x) + v'_3(-\sin x) &= \operatorname{cosec} x \end{aligned}$$

which follows from Extension: n th order.

Solving $v'_1 = \operatorname{cosec} x, v'_2 = -\cot x, v'_3 = -1$.

Integrating $v_1 = \log(\operatorname{cosec} x - \cot x), v_2 = -\log \sin x, v_3 = -x$.

$$\therefore y_p = \log(\operatorname{cosec} x - \cot x) - \cos x \log \sin x - x \sin x.$$

$\therefore$ Complete solution is $y = c_1 + c_2 \cos x + c_3 \sin x + \log(\operatorname{cosec} x - \cot x) - \cos x \log \sin x - x \sin x$.

► **Example 5.7.39** Solve by the method of variation of parameters:

$$\frac{d^2 y}{dx^2} - 3 \frac{dy}{dx} + 2y = \frac{e^x}{1 + e^x}.$$

[C.H. 1998]

Solution: To find C.F.

Auxiliary equation $m^2 - 3m + 2 = 0 = (m-1)(m-2)$, then $m = 1, 2$. C.F. is $c_1 e^x + c_2 e^{2x}$.

To find P.I.

First to show that e^x and e^{2x} are independent solutions.

Let $y_1 = e^x$, $y_2 = e^{2x}$.

The Wronskian of y_1 and y_2 is

$$W(y_1, y_2) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} e^x & e^{2x} \\ e^x & 2e^{2x} \end{vmatrix} = e^{3x} \neq 0.$$

Therefore, y_1 and y_2 are independent solutions.

Let the particular solution be

$$y_p = v_1(x)e^x + v_2(x)e^{2x}. \quad (5.7.39)$$

$$\therefore Dy_p = \{v_1(x)e^x + 2v_2(x)e^{2x}\} + \{v_1'(x)e^x + v_2'(x)e^{2x}\}$$

Choose $v_1(x)$ and $v_2(x)$ such that

$$v_1'(x)e^x + v_2'(x)e^{2x} = 0 \quad (5.7.40)$$

so that

$$\begin{aligned} Dy_p &= v_1(x)e^x + 2v_2(x)e^{2x} \\ \text{and } D^2y_p &= \{v_1(x)e^x + 4v_2(x)e^{2x}\} + \{v_1'(x)e^x + 2v_2'(x)e^{2x}\}. \end{aligned}$$

Now the given equation is

$$\begin{aligned} D^2y_p - 3Dy_p + 2y_p &= \frac{e^x}{1+e^x} \\ \text{or, } v_1'(x)e^x + 2v_2'(x)e^{2x} &= \frac{e^x}{1+e^x}. \end{aligned} \quad (5.7.41)$$

From (5.7.40) and (5.7.41)

$$\begin{aligned} v_1'(x)e^x + v_2'(x)e^{2x} &= 0 \\ v_1'(x)e^x + 2v_2'(x)e^{2x} &= \frac{e^x}{1+e^x}. \end{aligned}$$

$$\text{Solving } v_1'(x) = -\frac{1}{1+e^x}, \quad v_2'(x) = \frac{1}{e^x(1+e^x)}.$$

$$\text{Whereby } v_1(x) = \log(1+e^x) - x, \quad v_2(x) = \log(1+e^x) - x - \frac{1}{e^x}.$$

$$\text{Accordingly, } y_p = (e^x + e^{2x}) \log(1+e^x) - (x+1)e^x - xe^{2x}.$$

Complete solution is

$$y = C_1e^x + C_2e^{2x} + (e^x + e^{2x}) \log(1+e^x) - (x+1)e^x - xe^{2x}.$$

Ex. 1. Solve : $\frac{dx}{dt} - 7x + y = 0$

$$\frac{dy}{dt} - 2x - 5y = 0$$

Solution : The equations can be written in symbolic form as

$$(D - 7)x + y = 0$$

...(i)

$$-2x + (D - 5)y = 0$$

...(ii)

operating equation (i) by $(D - 5)$ and then subtracting (ii), from it we can have $\{(D - 5)(D - 7) + 2\} x = 0$
 or $(D^2 - 12D + 37) x = 0$
 which is a linear equation with constant coefficients. Auxiliary equation for which will be

$(m^2 - 12m + 37) = 0$, which gives $m = 6 \pm i$.
 So the value of x will be

$$x = e^{6t} (c_1 \cos t + c_2 \sin t).$$

Value of y can be obtained either by the elimination of x between (i) and (ii) or by the substitution of x in any one of the equations. Here it is being obtained by both methods.

y-by elimination

Multiplying equation (i) by 2 and operating (ii) by $(D - 7)$ and then adding, we get

$$\begin{aligned} \text{or} \quad & \{2 + (D - 5)(D - 7)\} y = 0 \\ & (D^2 - 12D + 37) y = 0 \end{aligned}$$

which gives $y = e^{6t} (A \cos t + B \sin t)$ (iv)

The arbitrary constants c_1 and c_2 obtained in the solution (iii) for x and A and B obtained in the solution (iv) for y are not independent. In fact A and B can be expressed in terms of c_1 and c_2 as follows :

Substituting (iii) and (iv) in equation (i), we get

$$e^{6t} (A - c_1 + c_2) \cos t + (B - c_1 - c_2) \sin t = 0$$

This equation must be identically satisfied and therefore

$$A = c_1 - c_2 \text{ and } B = c_1 + c_2$$

Hence the required solution for y is

$$y = e^{6t} [(c_1 - c_2) \cos t + (c_1 + c_2) \sin t].$$

y-by substitution

We have $x = e^{6t} (c_1 \cos t + c_2 \sin t)$

$$\begin{aligned} \text{so} \quad \frac{dx}{dt} &= 6 e^{6t} (c_1 \cos t + c_2 \sin t) \\ &+ e^{6t} (-c_1 \sin t + c_2 \cos t) \end{aligned}$$

Substituting above values of x and $\frac{dx}{dt}$ in equation (i), we can obtain

$$\begin{aligned} y = 7x - \frac{dx}{dt} &= [7 e^{6t} (c_1 \cos t + c_2 \sin t) \\ &- 6 e^{6t} (c_1 \cos t + c_2 \sin t) \\ &- e^{6t} (-c_1 \sin t + c_2 \cos t)] \end{aligned}$$

$$\text{or} \quad y = e^{6t} \{(c_1 - c_2) \cos t + (c_1 + c_2) \sin t\}$$

Ex. 2. Solve : $\frac{dx}{dt} + \frac{dy}{dt} - 2y = 2 \cos t - 7 \sin t$

$$\frac{dx}{dt} - \frac{dy}{dt} + 2x = 4 \cos t - 3 \sin t$$

Solution : Writing in symbolic form, we have

$$Dx + (D - 2)y = 2 \cos t - 7 \sin t \quad \dots(i)$$

$$(D + 2)x - Dy = 4 \cos t - 3 \sin t \quad \dots(ii)$$

Operating both sides of (i) by D and (ii) by $(D - 2)$, we obtain following equations

$$D^2x + D(D - 2)y = -2 \sin t - 7 \cos t$$

$$(D^2 - 4)x - D(D - 2)y = 2 \sin t - 11 \cos t \quad \dots(iii)$$

Adding equations (iii) and (iv), we have

$$(D^2 - 2)x = -9 \cos t \quad \dots(iv)$$

which on integration gives

$$x = c_1 e^{\sqrt{2}t} + c_2 e^{-\sqrt{2}t} + 3 \cos t$$

Now,

$$\frac{dx}{dt} = \sqrt{2} c_1 e^{\sqrt{2}t} - \sqrt{2} c_2 e^{-\sqrt{2}t} - 3 \sin t$$

substituting the values of x and $\frac{dx}{dt}$ in (ii), we get

$$Dy = c_1 e^{\sqrt{2}t} (2 + \sqrt{2}) + c_2 e^{-\sqrt{2}t} (2 - \sqrt{2}) + 2 \cos t$$

On integration

$$y = (1 + \sqrt{2}) c_1 e^{\sqrt{2}t} + (1 - \sqrt{2}) c_2 e^{-\sqrt{2}t} + 2 \sin t.$$

the constant of integration obtained in it will be zero because the values of x and y must satisfy the given equations (i) and (ii).

Ex. 3. Solve : $\frac{d^2x}{dt^2} - 3x - 4y = 0$

$$\frac{d^2y}{dt^2} + x + y = 0$$

Solution : In symbolic manner the given equation can be written as

$$(D^2 - 3)x - 4y = 0 \quad \dots(i)$$

$$x + (D^2 + 1)y = 0 \quad \dots(ii)$$

Operating both sides of (i) by $(D^2 + 1)$ and multiplying (ii) by 4, we obtain

$$(D^2 + 1)(D^2 - 3)x - 4(D^2 + 1)y = 0 \quad \dots(iii)$$

$$4x + 4(D^2 + 1)y = 0 \quad \dots(iv)$$

Adding (iii) and (iv), we obtain

$$(D^4 - 2D^2 + 1)x = 0$$

which is a linear equation with constant coefficients. Auxiliary equation in this case will be

$$(m^4 - 2m^2 + 1) = 0, \text{ which gives } m = \pm 1, \pm 1.$$

Thus the value of x will be given by

$$x = (c_1 + c_2 t) e^t + (c_3 + c_4 t) e^{-t}$$

Substitution of x in any of the equations will produce

$$2y = (c_2 - c_1 - c_2 t) e^t - (c_3 + c_4 + c_4 t) e^{-t}$$

These values of x and y taken together will constitute the whole solution.

x. 4. Solve the simultaneous equations :

$$\frac{dx}{dt} + 2x = t$$

$$\frac{dy}{dt} - 2y = \frac{1}{t}$$

Solution : The given equations can be written as

$$(D + 2)x = t \quad \text{.....(i)}$$

$$-2x + Dy = \frac{1}{t} \quad \text{....(ii)}$$

Here the value of x can be directly obtained by the integration of (i), as it does not contain variable y .

$$\text{We obtain } x = c_1 e^{-2t} + \frac{t}{2} - \frac{1}{4}$$

Substituting the above value of x in equation (ii), we get

$$Dy = 2c_1 e^{-2t} + t - \frac{1}{2} + \frac{1}{t}$$

$$\text{Integrating, } y = -c_1 e^{-2t} + c_2 + \frac{t^2}{2} - \frac{t}{2} + \log t$$

These two values for x and y constitute the whole solution.

► Example 6.5

Find the power series solution of the differential equation

$$\frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 + 2)y = 0 \quad (6.4)$$

in powers of x (that is, about $x_0 = 0$).

Solution. We have already observed that $x_0 = 0$ is an ordinary point of the differential equation (6.4) and that two linearly independent solutions of the desired type actually exist. Our procedure will yield both of these solutions at once.

We thus assume a solution of the form (6.6) with $x_0 = 0$. That is, we assume

$$y = \sum_{n=0}^{\infty} c_n x^n. \quad (6.10)$$

Differentiating term by term we obtain

$$\frac{dy}{dx} = \sum_{n=1}^{\infty} n c_n x^{n-1} \quad (6.11)$$

and

$$\frac{d^2 y}{dx^2} = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}. \quad (6.12)$$

Substituting the series (6.10), (6.11), and (6.12) into the differential equation (6.4), we

obtain

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + x \sum_{n=1}^{\infty} n c_n x^{n-1} + x^2 \sum_{n=0}^{\infty} c_n x^n + 2 \sum_{n=0}^{\infty} c_n x^n = 0.$$

Since x is independent of the index of summation n , we may rewrite this as

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} + \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+2} + 2 \sum_{n=0}^{\infty} c_n x^n = 0. \quad (6.13)$$

In order to put the left member of Equation (6.13) in the form (6.9), we shall rewrite the first and third summations in (6.13) so that x in each of these summations will have the exponent n . Let us consider the first summation

$$\sum_{n=2}^{\infty} n(n-1)c_n x^{n-2} \quad (6.14)$$

in (6.13). To rewrite the summation (6.14) so that x will have the desired exponent n , we first replace the present exponent $n-2$ in (6.14) by a new variable m . That is, we let $m = n-2$ in (6.14). Then $n = m+2$, and since $m = 0$ for $n = 2$, the summation (6.14) takes the form

$$\sum_{m=0}^{\infty} (m+2)(m+1)c_{m+2} x^m. \quad (6.15)$$

Now since the variable of summation is merely a "dummy" variable, we may replace m by n in (6.15) to write the first summation in (6.13) as

$$\sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2} x^n. \quad (6.16)$$

In like manner, letting $m = n+2$, the third summation

$$\sum_{n=0}^{\infty} c_n x^{n+2} \quad (6.17)$$

in (6.13) first takes the form

$$\sum_{m=2}^{\infty} c_{m-2} x^m. \quad (6.18)$$

Then replacing m by n in (6.18), the third summation in (6.13) may be written as

$$\sum_{n=2}^{\infty} c_{n-2} x^n. \quad (6.19)$$

Thus replacing (6.14) by its equivalent (6.16) and (6.17) by its equivalent (6.19), Equation (6.13) may be written

$$\sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2} x^n + \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=2}^{\infty} c_{n-2} x^n + 2 \sum_{n=0}^{\infty} c_n x^n = 0. \quad (6.20)$$

Although x has the same exponent n in each summation in (6.20), the ranges of the various summations are not all the same. In the first and fourth summations n ranges from 0 to ∞ , in the second n ranges from 1 to ∞ , and in the third the range is from 2 to ∞ . The common range is from 2 to ∞ . We now write out individually the terms in each summation that do *not* belong to this common range, and we continue to employ the

"sigma" notation to denote the remainder of each such summation. For example, in the first summation

$$\sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2}x^n$$

of (6.20) we write out individually the terms corresponding to $n=0$ and $n=1$ and denote the remainder of this summation by

$$\sum_{n=2}^{\infty} (n+2)(n+1)c_{n+2}x^n.$$

We thus rewrite

$$\sum_{n=0}^{\infty} (n+2)(n+1)c_{n+2}x^n$$

in (6.20) as

$$2c_2 + 6c_3x + \sum_{n=2}^{\infty} (n+2)(n+1)c_{n+2}x^n.$$

In like manner, we write

$$\sum_{n=1}^{\infty} nc_nx^n$$

in (6.20) as

$$c_1x + \sum_{n=2}^{\infty} nc_nx^n$$

and

$$2 \sum_{n=0}^{\infty} c_nx^n$$

in (6.20) as

$$2c_0 + 2c_1x + 2 \sum_{n=2}^{\infty} c_nx^n.$$

Thus Equation (6.20) is now written as

$$2c_2 + 6c_3x + \sum_{n=2}^{\infty} (n+2)(n+1)c_{n+2}x^n + c_1x + \sum_{n=2}^{\infty} nc_nx^n + \sum_{n=2}^{\infty} c_{n-2}x^n + 2c_0 + 2c_1x + 2 \sum_{n=2}^{\infty} c_nx^n = 0.$$

We can now combine like powers of x and write this equation as

$$(2c_0 + 2c_2) + (3c_1 + 6c_3)x + \sum_{n=2}^{\infty} [(n+2)(n+1)c_{n+2} + (n+2)c_n + c_{n-2}]x^n = 0. \quad (6.21)$$

Equation (6.21) is in the desired form (6.9). For (6.21) to be valid for all x in the interval of convergence $|x - x_0| < R$, the coefficient of each power of x in the left

member of (6.21) must be equated to zero. This leads immediately to the conditions

$$2c_0 + 2c_2 = 0, \quad (6.22)$$

$$3c_1 + 6c_3 = 0, \quad (6.23)$$

and

$$(n+2)(n+1)c_{n+2} + (n+2)c_n + c_{n-2} = 0, \quad n \geq 2. \quad (6.24)$$

The condition (6.22) enables us to express c_2 in terms of c_0 . Doing so, we find that

$$c_2 = -c_0. \quad (6.25)$$

The condition (6.23) enables us to express c_3 in terms of c_1 . This leads to

$$c_3 = -\frac{1}{2}c_1. \quad (6.26)$$

The condition (6.24) is called a *recurrence formula*. It enables us to express each coefficient c_{n+2} for $n \geq 2$ in terms of the previous coefficients c_n and c_{n-2} , thus giving

$$c_{n+2} = -\frac{(n+2)c_n + c_{n-2}}{(n+1)(n+2)}, \quad n \geq 2. \quad (6.27)$$

For $n = 2$, formula (6.27) is

$$c_4 = -\frac{4c_2 + c_0}{12}.$$

Now using (6.25), this reduces to

$$c_4 = \frac{1}{4}c_0, \quad (6.28)$$

which expresses c_4 in terms of c_0 . For $n = 3$, formula (6.27) is

$$c_5 = -\frac{5c_3 + c_1}{20}.$$

Now using (6.26), this reduces to

$$c_5 = \frac{3}{40}c_1, \quad (6.29)$$

which expresses c_5 in terms of c_1 . In the same way we may express each even coefficient in terms of c_0 and each odd coefficient in terms of c_1 .

Substituting the values of c_2, c_3, c_4 , and c_5 , given by (6.25), (6.26), (6.28), and (6.29), respectively, into the assumed solution (6.10), we have

$$y = c_0 + c_1x - c_0x^2 - \frac{1}{2}c_1x^3 + \frac{1}{4}c_0x^4 + \frac{3}{40}c_1x^5 + \cdots.$$

Collecting terms in c_0 and c_1 , we have finally

$$y = c_0(1 - x^2 + \frac{1}{4}x^4 + \cdots) + c_1(x - \frac{1}{2}x^3 + \frac{3}{40}x^5 + \cdots), \quad (6.30)$$

which gives the solution of the differential equation (6.4) in powers of x through terms in x^5 . The two series in parentheses in (6.30) are the power series expansions of two linearly independent solutions of (6.4), and the constants c_0 and c_1 are arbitrary constants. Thus (6.30) represents the general solution of (6.4) in powers of x (through terms in x^5).

Find a power series solution of the initial-value problem

$$(x^2 - 1) \frac{d^2 y}{dx^2} + 3x \frac{dy}{dx} + xy = 0, \quad (6.31)$$

$$y(0) = 4, \quad (6.32)$$

$$y'(0) = 6. \quad (6.33)$$

Solution. We first observe that all points except $x = \pm 1$ are ordinary points for the differential equation (6.31). Thus we could assume solutions of the form (6.6) for any $x_0 \neq \pm 1$. However, since the initial values of y and its first derivative are prescribed at $x = 0$, we shall choose $x_0 = 0$ and seek solutions in powers of x . Thus we assume

$$y = \sum_{n=0}^{\infty} c_n x^n. \quad (6.34)$$

Differentiating term by term we obtain

$$\frac{dy}{dx} = \sum_{n=1}^{\infty} n c_n x^{n-1} \quad (6.35)$$

and

$$\frac{d^2 y}{dx^2} = \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2}. \quad (6.36)$$

Substituting the series (6.34), (6.35), and (6.36) into the differential equation (6.31), we obtain

$$\sum_{n=2}^{\infty} n(n-1) c_n x^n - \sum_{n=2}^{\infty} n(n-1) c_n x^{n-2} + 3 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=0}^{\infty} c_n x^{n+1} = 0. \quad (6.37)$$

We now rewrite the second and fourth summations in (6.37) so that x in each of these summations has the exponent n . Doing this, Equation (6.37) takes the form

$$\sum_{n=2}^{\infty} n(n-1) c_n x^n - \sum_{n=0}^{\infty} (n+2)(n+1) c_{n+2} x^n + 3 \sum_{n=1}^{\infty} n c_n x^n + \sum_{n=1}^{\infty} c_{n-1} x^n = 0. \quad (6.38)$$

The common range of the four summations in (6.38) is from 2 to ∞ . We can write out the individual terms in each summation that do *not* belong to this common range and thus express (6.38) in the form

$$\begin{aligned} \sum_{n=2}^{\infty} n(n-1) c_n x^n - 2c_2 - 6c_3 x - \sum_{n=2}^{\infty} (n+2)(n+1) c_{n+2} x^n \\ + 3c_1 x + 3 \sum_{n=2}^{\infty} n c_n x^n + c_0 x + \sum_{n=2}^{\infty} c_{n-1} x^n = 0. \end{aligned}$$

Combining like powers of x , this takes the form

$$\begin{aligned} -2c_2 + (c_0 + 3c_1 - 6c_3)x \\ + \sum_{n=2}^{\infty} [-(n+2)(n+1) c_{n+2} + n(n+2) c_n + c_{n-1}] x^n = 0. \quad (6.39) \end{aligned}$$

For (6.39) to be valid for all x in the interval of convergence $|x - x_0| < R$, the coefficient of each power of x in the left member of (6.39) must be equated to zero. In doing this, we are led to the relations

$$-2c_2 = 0,$$

$$c_0 + 3c_1 - 6c_3 = 0,$$

(6.40)

(6.41)

and

$$-(n+2)(n+1)c_{n+2} + n(n+2)c_n + c_{n-1} = 0, \quad n \geq 2.$$

(6.42)

From (6.40), we find that $c_2 = 0$; and from (6.41), $c_3 = \frac{1}{6}c_0 + \frac{1}{2}c_1$. The recurrence formula (6.42) gives

$$c_{n+2} = \frac{n(n+2)c_n + c_{n-1}}{(n+1)(n+2)}, \quad n \geq 2.$$

Using this, we find successively

$$c_4 = \frac{8c_2 + c_1}{12} = \frac{1}{12}c_1,$$

$$c_5 = \frac{15c_3 + c_2}{20} = \frac{1}{8}c_0 + \frac{3}{8}c_1.$$

Substituting these values of $c_2, c_3, c_4, c_5, \dots$ into the assumed solution (6.34), we have

$$y = c_0 + c_1x + \left(\frac{c_0}{6} + \frac{c_1}{2}\right)x^3 + \frac{c_1}{12}x^4 + \left(\frac{c_0}{8} + \frac{3c_1}{8}\right)x^5 + \dots$$

or

$$y = c_0(1 + \frac{1}{6}x^3 + \frac{1}{8}x^5 + \dots) + c_1(x + \frac{1}{2}x^3 + \frac{1}{12}x^4 + \frac{3}{8}x^5 + \dots). \quad (6.43)$$

The solution (6.43) is the general solution of the differential equation (6.31) in powers of x (through terms in x^5).

We must now apply the initial conditions (6.32) and (6.33). Applying (6.32) to (6.43) we immediately find that

$$c_0 = 4.$$

Differentiating (6.43), we have

$$\frac{dy}{dx} = c_0(\frac{1}{2}x^2 + \frac{5}{8}x^4 + \dots) + c_1(1 + \frac{3}{2}x^2 + \frac{1}{3}x^3 + \frac{15}{8}x^4 + \dots). \quad (6.44)$$

Applying (6.33) to (6.44) we find that

$$c_1 = 6.$$

Thus the solution of the given initial-value problem in powers of x (through terms in x^5) is

$$y = 4(1 + \frac{1}{6}x^3 + \frac{1}{8}x^5 + \dots) + 6(x + \frac{1}{2}x^3 + \frac{1}{12}x^4 + \frac{3}{8}x^5 + \dots)$$

or

$$y = 4 + 6x + \frac{11}{3}x^3 + \frac{1}{2}x^4 + \frac{11}{4}x^5 + \dots$$